# MILL TEST CERTIFICATE

AXIS
PIPE AND TUBE

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| | | | |
|---|---|---|---|
| **Customer:** | | **Sales Order:** | **Invoice No.:** |
| DNOW L.P. | | 0630005623 | 1600052358 |
| PO BOX 40985 | | **PO:** | **Date of Issue:** |
| Houston,TX,77240-0985 | | 26708945 | 03/19/2025 |
| 281-823-4700 | | | |
| **Certificate No.:** | 1600052358-3 | | |
| **Product Description:** | 5L 10.750 0.500 54.79# X52BDRH5BW-20P2 | | |
| **Specification:** | 5L 46th Edition May 2019 | | |

| Heat Number | Lot Number | Mat Number | Steel Making / Coil Rolling |
|---|---|---|---|
| A416856 | LT25003005 | 333870 | Steel Dynamics Columbus, LLC , Columbus - MS - US / Steel Dynamics Columbus, LLC , Columbus - MS - US |

**Mechanical Properties**

| Tensile Test - Strip Specimen Gauge Length 2" | | | | | | | | Temp (F) -20 | | | | CVN | | | | Temp (F) -20 | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Body (ft-lbf) | | | | Shear Area (%) | | | | Weld (ft-lbf) | | | | Shear Area (%) | | | | | | | |
| Loc. / Orien | Size | 0.5%EUL Y.S(psi) | U.T.S. Body (psi) | U.T.S. Weld (psi) | Y/T Ratio | E.L. (%) | Hardness (HRB) | 1 | 2 | 3 | Avg | 1 | 2 | 3 | Avg | 1 | 2 | 3 | Avg | 1 | 2 | 3 | Avg | Size | Loc. / Orien | Acceptance Criteria (ft-lbf)(AVG) |
| T180° | 1 1/2" | 67800 | 73300 | 78100 | 0.93 | 40.4 | 82 | 202 | 162 | 191 | 185 | 100 | 90 | 100 | 97 | 198 | 199 | 190 | 196 | 100 | 100 | 100 | 100 | 10x7.5 mm | 3T90 | 15 |
| | | | | | | | | Lateral Expasion (in) | | | | 0.101 | UB | UB | 0.101 | Lateral Expasion (in) | | | | UB | UB | UB | UB | | | |

| Analysis (H/P) | Chemical Composition (Max %) | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | C | Si | Mn | P | S | Ni | Cr | Mo | Cu | V | Al | Nb | Ti | B | Ca | N | CEpcm (API Max. A.C.) | CE Results |
| H | 0.06 | 0.210 | 0.81 | 0.020 | 0.003 | 0.06 | 0.09 | 0.020 | 0.13 | 0.002 | 0.022 | 0.026 | 0.015 | 0.0003 | 0.0018 | 0.0069 | 0.25 | 0.12 |
| P | 0.06 | 0.220 | 0.80 | 0.017 | 0.002 | 0.05 | 0.08 | 0.020 | 0.13 | 0.003 | 0.020 | 0.025 | 0.015 | 0.0002 | 0.0017 | 0.0086 | 0.25 | 0.12 |
| P | 0.06 | 0.221 | 0.80 | 0.018 | 0.002 | 0.05 | 0.08 | 0.020 | 0.13 | 0.003 | 0.020 | 0.025 | 0.015 | 0.0002 | 0.0019 | 0.0093 | 0.25 | 0.12 |

**Following Controls / Tests have been satisfactorily performed:**
1.- Hydrostatic Testing: 4,860.00 psi Pressure for 5.00 seconds
2: Ultrasonic Tested to N10 - passed
3: Seam Annealed min 1,600 F
4: Flattening Test Passed

**Remarks:**
1: This material meets:Grades ASTM A53/A53M-22 Grade B
API 5L HFW PSL2 X52M,
2: This product meets the requirements of ASME BPVC.II.A Specification SA-53B-2023
3: Made and Melted in USA
4: Date of Manufacture 01/2025
5: This product is silicon and/or aluminum fully killed.
6: Certificate meets the requirements of ISO 10474:2013 (Inspection Certificate 3.1.B)
7: This material meets the requirements of NACE MR0175-2021 / ISO 15156:2020

**Acronyms:**
L = Longitudinal
Y.S. = Yield Strength
E.L. = Elongation
P = Product Analysis
UB = Un-Broken
T = Transverse
U.T.S. = Ultimate Tensile Strength
H = Heat Analysis
A.C. = Acceptance Criteria
N/A = Not Applicable

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

Mahir Haroa
Quality Manager

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FORACA-AXIS-X2

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